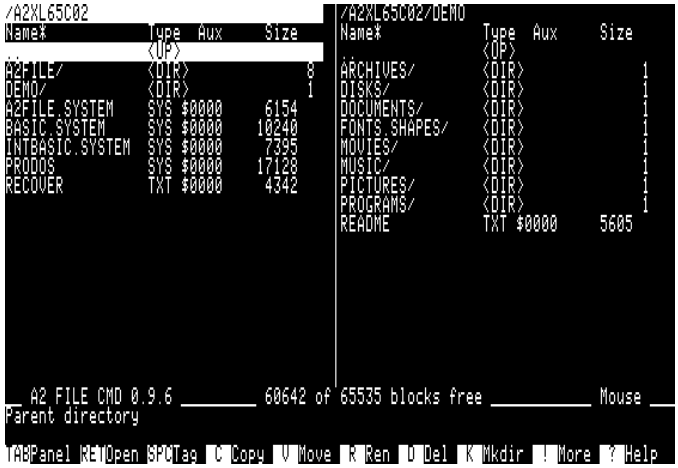
A2 FILE CMD

User guide · 0.9.9

DOS3.3: the essentials · ProDOS: the complete workspace

From 48 KB and 40 columns to 128 KB and 80 columns



A2 File Cmd ProDOS XL 0.9.6 in 80 columns: two panels captured in POM2

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A2 File Cmd DOS3.3

Two panels, a source and a destination: DOS3.3 distils A2 File Cmd into selecting, reading, editing and verified copying. It uses an **Apple II+ with 48 KB**, NMOS 6502, 40 columns and **two Disk II drives on the boot controller**. The following chapters extend this workflow with ProDOS folders and tools.

Boot `A2FILECMD-DOS3.3-0.9.9.dsk`, or use **BRUN A2FC** from DOS 3.3. Both panels start on the boot disk; each remembers drive, selection and scroll. Catalogs hold up to 105 files and show 19 rows. ? opens help.

Keys	Action
TAB/Ctrl-I or 1; / or 4	Switch panels; change active drive.
Up/Down or I/K; Left/Right	Previous/next file; previous/next page. [/] first/last.
Return or 2; T/H/G; B	Open; force text/hex/hi-res; confirm BRUN.
Space; Ctrl-T/N; *	Tag one; all/none; invert tags.
C or 3; N/E; D/R/L	Copy; new/edit text; delete/rename/lock.
F	Format the active drive (not the boot one) with DOS copied from the boot disk.
Ctrl-R or 5; =	Reread both panels; same disk opposite.
? or 6; Q or 7; Escape	Help; confirm quit; leave preview/help/editor.

Questions use **Y** to confirm and **N/Escape** to cancel. Other keys do not confirm. Escape in the browser has no effect: DOS has no parent directories. After changing a disk, reread with Ctrl-R. Prompts and the result occupy the footer; read them before inserting another disk.

Copy and edit

1. Put the panels on different drives using TAB and `/`.
2. Select/tag the sources and press C. Check `COPY name?` or the marked count.
3. Confirm Y, then wait for the progress bar and final result. **Once started, the disk-writing operation cannot be interrupted from the keyboard.**

Copies preserve DOS names, type, lock flag and all data-sector bytes. Existing names are skipped. Destination sectors are reserved first, written and read back, then the catalog entry is published. An uncertain write stops the batch; the source is neither written nor deleted. There is no single-drive copy by swapping disks.

N/E uses the same exclusive writer. E accepts at most **8 KB** and refuses larger files rather than truncating them. It always saves to a new name. A save that does not land (name taken, write-protected or full disk, read error, N or Escape at the question) keeps the text and returns to the editor: Ctrl-S tries again, and only Escape then N Abandon gives the text up. R refuses an existing name or locked file. D skips locked files; L toggles the cursor's lock, or unlocks a tagged set if any member is locked, otherwise locks the set.

Preview and limits

Return identifies text, hi-res or a binary program from content and DOS headers; a program asks `BRUN NAME?`. T/H previews only the first 256 stored bytes; G shows the first 8 KB as hi-res. The hex preview shows eight bytes a row with their characters, one half of the sector at a time: Left/Right (or -/+, </>) switch between bytes 00-7F and 80-FF. A valid 8 KB BIN load at $2000/$4000 is a picture; unsupported or unsafe program headers fall back to hex. BRUN leaves A2FC. Pinball Construction Set `*.PB` tables open in hexadecimal with Return or B; they are data even when their DOS header looks executable.

Only standard 35-track, 16-sector DOS 3.3 is supported. Sparse/inconsistent chains, reserved-track data, 13-sector, 40-track and protected formats are refused. Before a write, the catalog entry must still match the panel's identity; a changed disk is refused, though a byte-identical twin cannot be distinguished. Physical write protection is checked. After an uncertain error, reserved space is retained and further writes are blocked for that run: preserve and inspect a copy of the disk. Readback cannot repair metadata damaged by a power failure.

From DOS3.3 to ProDOS

ProDOS builds on the same two-panel workflow: folders, attributes, archives, documents, pictures, music and disk tools. It needs an **Apple IIe, //c or IIgs with 128 KB and 80 columns**. This is a separate program: its commands, plugins and recovery filenames do not apply to the DOS3.3 edition introduced first.

Four self-contained images

Image	Choose it for
`A2FILECMD-DOS3.3-0.9.9.dsk`	Standalone DOS 3.3 file manager for Apple II+ and two Disk II drives.
`A2FILECMD-PRODOS-800K-0.9.9.po`	Selected practical tools, BASIC runtimes and free space for files; any IIe.
`A2FILECMD-PRODOS-XL-0.9.9.2mg`	Complete 32 MB ProDOS edition with tools and demonstrations; any IIe.
`A2FILECMD-PRODOS-XL-65C02-enhanced-0.9.9.2mg`	Complete XL for an enhanced IIe, //c or IIgs: MouseText and optional mouse support.

The choice is **enhanced or not**. The enhanced XL needs an enhanced IIe, a //c or a IIgs; every other ProDOS image runs on any IIe, enhanced or not. An enhanced machine always has the 65C02 the enhanced XL is compiled for: the launcher checks machine, processor and memory before starting, so a hand-modified machine is refused rather than crashed. Mouse hardware is optional.

DOS3.3 uses DOS-sector-order `.dsk`; 800K uses ProDOS-order `.po`. XL uses `.2mg`. Renaming an extension does not convert an image. With all release images and the PDF in the same directory, check their downloads:

```
sha256sum -c SHA256SUMS-0.9.9.txt
```

Install and start

Boot the selected image, or launch `A2FILE.SYSTEM` from a ProDOS selector. When installing elsewhere, keep `A2FILE.SYSTEM` beside the complete `A2FILE/` directory and copy `RECOVER` too. Keep the program and its native plugins from the same build and CPU edition; do not mix them.

The 800K edition includes 94 overlays selected for everyday file management, disk imaging, recovery, archives and common documents, pictures and music. It includes SMALLDOS, Pascal/CPM readers and writers, BASIC/assembly readers, AppleWorks, Apple Writer, PFS File/Plan, VisiCalc, HGR/DGR, MacPaint, Print Shop and the music players, plus both BASIC runtimes and the movie players. The explicit inventory is in `config/800k.mk`. The current image leaves 278 blocks (139 KiB) free; builds require at least 160 free blocks. XL retains all tools, including niche document, graphics and source formats. Only XL includes `DEMO/`: one example of every kind of file A2 File Cmd opens, in DOCUMENTS, PICTURES (an HGR album, and in CARDS two colour cards in every picture format), MOVIES (Fantavision, and a Take 1 movie on its DOS 3.3 disk), MUSIC, ARCHIVES, DISKS, PROGRAMS and FONTS.SHAPES, and a README saying what is where.

Before working

Use copies of valuable disks. Read each target name before confirming an operation. Keep space for both old and new versions during replacement. Do not remove a disk until the operation reports its result.

Some tools use auxiliary memory and rebuild **/RAM empty**. When /RAM holds files, A2FC asks before that use and explicitly warns that **ALL /RAM files will be lost**; an empty /RAM has nothing to lose and is used without a question. Copy the files elsewhere first; declining preserves them. A key typed before a question appears is dropped, never taken as its answer. This applies to affected picture viewers, the PT3 player (GROUiK's engine; a declined question plays with pt3_lib instead), large VisiCalc worksheets, ShrinkIt, disk-image operations, Disk II formatting, NIBCOPY and full large-volume checks. Ordinary text readers and the other music players preserve /RAM. Ctrl-Reset quits A2FC to ProDOS; pressed inside a tool that had leave to use auxiliary memory, it also rebuilds /RAM empty on the way out.

Keys and panels

TAB changes the active panel. The highlighted row is the cursor; ∗ marks a tagged entry and L a locked one. Directories end in /. The header's star shows the sort order. Messages and progress appear below the panels.

Keys	Action
Up/Down; Left/Right, < / > or - / +	Move one entry; move one page. [/] goes to first/last.
Return / Escape / /	Open; parent directory; list volumes. Return detects and opens the appropriate reader; X launches programs with confirmation.
TAB / = / Ctrl-R	Other panel; same directory opposite; reread both panels.
Space / * / Ctrl-T / Ctrl-N	Tag one; invert tags; tag all; clear tags.
S / M / ' then initial	Next sort (name, size, type); tag files missing opposite or of another size/date; jump to a name.
C / V / D	Copy; move; delete. Use tags, or the cursor when none are tagged.
R / K / A / L	Rename; new directory; type/auxtype; lock/unlock.
T / H / I / E / X	Text; hex; picture; edit; run after confirmation.
W / F / ! / ? / Q	Disk images; format; tools menu; help; quit after confirmation.
1 … 0	Activate the corresponding bottom key-bar button.

! lists tools by category. The menu holds up to **128 commands**, enough for all 116 visible commands in XL. Names are complete; descriptions show their first 27 characters. **Images** includes MVMOVIE, DGMAGI, PCSVW, PURPLE, PSBORDER, PSFONT and DOSFONT. Select the file in the panel before choosing its reader. The menu itself uses MAIN only; readers that need AUX retain their RAM-loss confirmation.

Names start with a letter, then letters, digits or periods, up to 15 characters. Directories can be tagged; **..** cannot. A tool or viewer that takes over the screen gives the tags back only to a panel that returns with the same entries; a panel whose contents changed meanwhile comes back untagged. Returning to a parent selects the child you left if it still exists. A directory of more than 139 entries is read in unsorted windows of 139, in disk order (the header gives the window's first entry, then `+ disk order`); continue through the window edge to reach later entries. A larger folder inside a disk image shows only its first 139 entries, in disk order, and is not paged: the rest are out of reach there.

Enhanced mouse support lets you select a row, click it again to open, click a path to go up, a header to sort or a key-bar button to act. Prompts, viewers and the editor use the keyboard. On quit, A2FC saves panel paths, sorting and active side; a failed preference save lets you cancel quitting.

Copy, move and edit

1. Open the destination in the other panel. In the source, clear unwanted tags with **Ctrl-N**, then select or tag the entries to transfer.

2. Press **C** or **V**. For an existing target: **O** overwrites, **S** skips, **A** overwrites all and **N** overwrites none. Check the destination first.

3. Wait for **Copying** and **Verifying**, then read the result. Move removes a source only after its copy is verified. Existing directories are filled in; a directory is never copied into itself, into one of its descendants or onto one of its ancestors. With both panels on one directory, C and V refuse: "Both panels show the same directory." Paths from GOTO and A2FILE.CFG are kept in capitals, as ProDOS reads them, so `/workhd/dir` and `/WORKHD/DIR` are one directory.

Copy preserves type and auxtype. Replacements retain the old target during verification. **Escape** interrupts; completed work remains. If cleanup fails, a temporary can remain: use the recovery procedure before retrying. A source tree that is too deep, or holds a path longer than 63 characters, is refused whole before its first file is copied or deleted; process smaller subdirectories rather than bypassing the limit.

! → **Files** → **MOVE** can relocate a single entry within a volume without copying its data blocks. A marked batch, directories included, first reserves `A2MOVE.LST` and stops on error or Escape. A directory bound for another volume is copied, verified and only then deleted, as with V; cross-volume copies are verified before source deletion. Keep any retained work list for review.

E edits up to **5,104 bytes**, with CR line endings; a larger file is refused (`Too big for the editor (5 KB).`), never cut short, and so is one that grew since the panel read its size (`Open failed.`: press Ctrl-R and try again). Loading strips the high bit and long lines do not wrap. On a directory or `..`, E creates text. Arrows move; Delete/Ctrl-D erase left/right; Ctrl-A/E goes to line start/end; Ctrl-P/N changes page; Ctrl-T/B goes to text start/end. Return splits a line; Tab inserts four spaces. Escape opens **S** save, **X** save/exit, **Q** abandon (confirm changes), or Escape to continue. Saving preserves type/auxtype and uses verified `A2FC.EDIT` plus `A2FC.ED.BAK`; existing recovery files block saving.

Recover after an incident

Every ProDOS disk includes **RECOVER** at its root. Open it with Return; the ? help screen points to it. Recovery begins by preserving evidence, not by retrying the failed operation or deleting the files it left behind.

Preserve, identify, recover

1. Stop and record the exact error, source path and intended destination. Make a disk image or duplicate before changing anything. Work on the duplicate and recover onto a separate healthy disk. If reads are unreliable and no independent copy is possible, stop and seek help.

2. Keep the target, temporary and backup. A name, date or matching size does not prove completeness. A generic `A2FC.BAK` may belong to an earlier operation: if its original target is unknown, preserve it without guessing.

3. On the **duplicate only**, rename one identified candidate to an unused `RECOV.OLD` or `RECOV.NEW`. This is necessary for `A2FC.COPY`, which C cannot copy under that reserved name. Choose another name if it is already taken.

4. Put a fresh recovery directory in the other panel, clear source tags with **Ctrl-N**, select the candidate and press **C**. Refuse overwrite. Wait for copying **and verification**; an error means recovery is not confirmed.

5. Check the recovered content with its reader, or compare all bytes against a known good source. Do not execute an unknown program to test it. Only after validation, give the recovered copy its intended name in an empty directory. Keep the disk image and other candidates until satisfied.

Recognize the files

Operation	New candidate	Previous version
Copy	`A2FC.COPY`	`A2FC.BAK`
Editor	`A2FC.EDIT`	`A2FC.ED.BAK`
SYNC	`A2FC.SYNC`	`A2FC.BAK`
Text / image conversion	`TXTCONV.TMP` / `IMGCONV.TMP`	`A2FC.BAK`
DOS image write	`A2FC.DOS`	`A2FC.BAK`
Preferences in A2FILE/	`A2FILE.TMP`	`A2FILE.BAK`
GOTO in A2FILE/	`GOTO.TMP`	`GOTO.BAK`

Target plus backup can mean installation succeeded but cleanup failed. **Backup without target** can mean restoration is needed. A **temporary** can be incomplete or verified but not installed. Preserve the candidates until checked; absence of a backup does not prove an interrupted write completed. A reported successful save with a retained backup is distinct from a failed save. Read the exact message and validate the saved target before cleanup.

For a partial move, compare both sides file by file. **A2MOVE.LST is a work list, not file contents**: do not replay, edit or delete it as a recovery shortcut. Copy missing files into a fresh directory and retain remaining sources. Extraction may leave partial output under its normal filename.

Keep power on for files in **/RAM**; do not reboot or accept AUX reclamation. A2FC does not infer original names, automatically resume interrupted work or promise survival of a power cut during physical metadata writes. More failure cases are documented in Data safety.

Read documents and pictures

Text and documents

T reads text; **H** shows offsets, bytes and characters. Space/Return/ Right/Down advances; B/Left/Up goes back; Escape returns. TEXT breaks long lines at column 80 and remembers 80 page starts; **R** restarts it. The status line of TEXT, the BASIC lister and the AppleWorks reader says `(end)` at the end of the file, `(READ ERROR)` when a read failed, and `(page limit)` on the eightieth page of a file that goes on. Use **MDVIEW** for word-wrapped text and Markdown, with 64 previous pages and no forward limit. It also reads extracted Magic Window `.MW` documents. Teach extended files are unsupported. Return on an Epistole document (text opening on a _ command) or a Papyrus or HomeWord one (opening on a $FF code; the others through **!**) opens **DOCVIEW**, laid out as printed: margins, indents, centring, bold in inverse, mail-merge variables. French ISO 646-FR accents show unaccented; **A** shows them as stored, for a French character set. Epistole's calculations are computed as Epistole prints them, on the Applesoft ROM's arithmetic: `#:X=…]` sets a variable, `#:?…]` shows a value with `_ND` decimals (2 by default), a decimal comma, its comma on the `_TD` tab (a tab beyond the right margin: the number starts the next row), and no minus sign when there are decimals, as Epistole prints it. What cannot be computed (a division by zero, a number beyond 1E38, a variable typed at print time, `#*X=]`) stays as written, in inverse. Headers (`_EN`) and footers (`_DB`, `%$` the page number) appear at each `_SP` page break and the footer again at the end; DOCVIEW does not repeat them at the printer's own page ends. Page numbers go to 255: the pages of a longer document all read 255 from there. It reads documents up to 64 KB and remembers 16 pages back. **H** jumps to the oldest page still remembered; **L** returns to the most advanced page already visited. Both retain history, layout, calculations and accent mode; they do not scan unvisited pages. For example, after visiting 20 pages, H goes to page 5 and L to page 20. **R** restarts the whole document and resets both destinations to page 1. Bank Street Writer documents (a BIN at $0840 or $63D0 from DOS 3.3, at 0 from ProDOS, high-bit text) open in DOCVIEW too, with Return: paragraphs wrapped, centred lines centred, tabs of four columns; the text ends at its first $00, the rest of the file being the program's leftovers. Extract them from a DOS 3.3 disk with **C** first. HEX uses **G** for a seven-digit offset and **R/E** for first/ last page.

Logo procedures open as text with Return: Apple Logo's (DOS 3.3 text files) as any text, Terrapin Logo's (`NAME.LOGO`, a BIN at $2000) by their suffix. T lists Applesoft, Integer BASIC and Business BASIC (`.BA3`) without running them. T/Return reads AppleWorks word-processing text, omitting layout commands. Return on an AppleWorks database or spreadsheet opens **AWDATA**: Space/Down and B/Up page, Left/Right moves columns, R restarts, and F switches between a sheet and cell details. Spreadsheet **L** jumps to the furthest indexed page of the current view, keeping the column window. This can include the next page prepared by the reader; it does not scan the whole sheet. **B** returns a page, **R** returns to the top left, and **F** changes view and restarts its index. Database **L** jumps to the last readable record using the checked walk and existing checkpoints. B goes back from there, R returns to the first, and TAB's category selection is retained. The help bar groups these ends as R/L. Database TAB shows categories 23–30. Limits are 2,688 records or 882 spreadsheet rows; displayed formulas/results are the saved ones, not a recalculation. Formats, reports and window settings are omitted.

Return on a VisiCalc worksheet (a `/SS` file: text whose first line is a cell such as `>B3:`; extract it from a DOS 3.3 disk with **C** first) opens **VISICALC**, which recalculates the sheet and shows it as VisiCalc does on loading it. A worksheet holds only what was typed, so the overlay computes every formula once, in the sheet's order (by columns or rows), with VisiCalc's own decimal arithmetic: twelve digits, truncated, operators from left to right (`2+3*4` is 20); a formula that refers to one not yet computed shows ERROR, as in VisiCalc until **!** was pressed. Cells appear in their columns with VisiCalc's formats (`/F$`, `/FI`, `/FL`, `/FR`, `/F*`, its general format and >>> overflow), the cursor cell in inverse, its name and contents on the first row. The arrows move the cursor, Space/Return and B go a screen down and up, > and < a screen of columns, R back to A1, Escape returns. A large sheet takes time to read and recalculate -- minutes when hundreds of formulas each add up a long range -- while the activity cell turns: **Escape** stops it there and returns to the panels (`Stopped.`). No other key is read meanwhile; one typed then is taken when the sheet shows. Powers and `@SQRT`, `@LN`, `@SIN`... are computed by the Applesoft ROM, whose last digits may differ from VisiCalc's own. A sheet of up to about 240 numbers and formulas fits in main memory; a larger one, up to 4,064, uses the auxiliary memory, which rebuilds /RAM afterwards and says so (also after Escape); A2 File Cmd asks first when /RAM holds files (No: the sheet is refused, /RAM untouched). Read only; VISICALC.BIN, beside the overlay, holds its other parts. Text that starts with > but is no worksheet is refused: **T** shows it as text. The format is described in VisiCalc worksheets.

Pictures: Return or I

Return and I select the same recognized picture formats on both CPU editions. Explicit packed types take priority; signatures identify DGR and HGRR/DHRR. An identification read/close error stops opening. **H** explicitly selects hex. 800K includes the common picture readers; specialized formats require **XL**.

Picture or font	Identification / tool
Raw HGR / DHGR, HGRR / DHRR	IMAGE; raw pages or version 1 RLE. Dazzle Draw pictures are raw DHGR.
Movie Maker	IMAGE; `.BKG` backgrounds are raw pages, `.SHP` shape sheets (BIN $1DF0, 8,720 bytes) a page after a 528-byte header.
KoalaPad (Micro-Illustrator)	IMAGE; `PICTR.*` are raw pages, BIN aux $4000, 8,184 bytes once copied from the DOS 3.3 data disk with **C**. C64/Atari Koala files are not Apple pictures: hex.
Lo-res / DGR, text screens	DGRVIEW; DGR signature or BIN/FOT with aux $0400 (a saved screen page: lo-res picture or text, told apart by content). I can ask for a raw sprite width.
Extasie / Arlequin	EXTASIE ($F2); ARLEQUIN ($F8 or imported ProDOS BIN with its `gs` signature).
MacPaint	MACPAINT; `.MAC`/`.PNTG`, or renamed ProDOS BIN/typeless MacBinary with the `PNTG` type field.
Packed FOT / LZ4FH	PACKFOT ($08/$4000 or $4001); LZ4FH ($08/$8066).
Packed 816/Paint	PAINT816; BIN with aux $E001 or $E002.
Purplesoft	PURPLE; matching `.FOT01` and `.FOT02` in one directory.
Print Shop	PRINTSHOP: 572/576-byte BIN clip art. PSBORDER: 144/148-byte BIN borders, aux $4800/$5800/$6800/$7800. PSFONT: BIN fonts, aux $6000/$5FF4; ProDOS $F5 borders/fonts aux $2000/$1000.
The Newsroom	NEWSROOM; photos `PH.*` and banners `BN.*`, BIN aux $4000 (copied from DOS 3.3).
Graphics Magician (Penguin, 1982-1984)	GMAGIC; a BIN whose first bytes are picture commands (no header, no name convention), Return only.
MGTK or hi-res fonts	FONTVIEW; type $07, a BIN of 768 or 1,024 bytes named `.SET` (DOS Tool Kit HRCG sets) or `.FONT` (Beagle's), or select the tool for an untyped font. DOS BIN `.SET`/`.FONT` fonts of these lengths also open directly through Return/I on 800K/XL.
Applesoft shapes	SHAPES; `.SHAPE` or choose the tool directly.

NEWSROOM: **H** mirrors horizontally, **V** vertically, and **I** inverts black and white. These act on the full 280×192 loaded preview, including its background. Repeat a command to undo it; combine H/V/I freely. Only visible HGR pixels change: screen holes, source files and AUX `/RAM` are preserved. Left/Right opens the next picture in its original orientation and colors. These controls apply to the ProDOS NEWSROOM viewer; direct DOS panels/layouts have their own controls.

Raw HGR accepts any size from **8,184 to 8,199 bytes** (a page eight bytes short, KoalaPad's re-saved 8,191, Terrapin Logo's 8,194); DHGR accepts **16,376 to 16,391**, auxiliary plane first: the one-file layout Dazzle Draw saves (BIN, aux $2000, 16,384 bytes), so its pictures and slide-show disks open as they are. Three real pictures match the original `DD.PICLOADER` byte for byte and in rendered screen output on both CPUs (qualification). Left/Right browses the previous/next file handled by the same viewer, including across directory windows (not inside a disk image). Right on the last one goes round to the first; Left on the first does nothing. **S** starts a slideshow in every picture viewer: the next picture comes by itself after about ten seconds (at 1 MHz; faster on an accelerated machine), round the folder without end. **S** again, or any other key, stops it; that key then acts as usual (Escape returns, an arrow moves). A picture alone in its folder is shown again. A picture that cannot be read ends the slideshow with its message. **Text screens.** A saved page of the screen (BIN at $0400, typically 1,016 to 1,024 bytes; 976 when the last row was not saved; more than 1,024 is read as 80 columns, auxiliary half first) is either a lo-res picture or a text screen: a title, crack or menu screen. DGRVIEW counts its visible bytes: a text screen has at least a quarter of spaces ($A0, inverse $20, $E0) and more spaces than "solid" bytes (two equal nibbles, the flat areas of a picture). On 77 real text screens and 45 real lo-res pictures from Asimov and French Touch, none was misread. The screen is shown by the Apple's own text mode: 40 columns in the primary set (flashing characters flash), 80 in the alternate set (MouseText). **T** reads the same bytes the other way (text or lo-res), **A** switches the character set; neither writes the screen again. Only the 960 visible bytes of each half are written: the screen holes keep the cards' state, and 80 columns write AUX $0400-$07FF, the 80-column text page, which /RAM never uses. Two separate 1,023-byte halves of an 80-column screen (as some disks keep them) are shown one at a time in 40 columns. While Left/Right looks for the next picture, the activity cell may appear for a moment at the bottom right of a text screen.

Loading shows the filename; Escape returns to the panels. Ordinary small BIN files open in hex with Return; I can treat a suitable one as a sprite. A viewer's consent to use AUX (see Before working) lasts only for that browsing session; leaving and reopening asks again.

NRCLIP (`!` menu, Images) turns a commercial Newsroom clip-art disk (the program's own and Clip Art Collections 1-3, which hold no files) into pictures. Open the disk in the active panel (a `.DSK`/`.DO`, a DOS-order `.2MG`, or the real floppy) and a ProDOS directory in the other one: each page is drawn on the hi-res screen, then saved there as a BIN $2000 of 8,192 bytes (a raw HGR page) named after the page, the way DOS 3.3 names are copied (`MEN 1` gives `MEN.1`). A name already there is skipped, never overwritten; a damaged page is skipped; a failed write removes its file and stops; Escape stops between pages. The message line counts the pages saved, already there and damaged. The clip-art disk is only read, and /RAM is kept.

GMAGIC draws The Graphics Magician's pictures (Penguin Software, 1982-1984, used by the adventure games of Penguin/Polarware: Transylvania, The Quest, Ring Quest...) the way the original routines do, on the visible hi-res page: lines, flood fills with the 108 patterns, brush stamps and, in the 1984 version, text. A picture is a list of drawing commands with no header: Return opens a BIN whose first bytes are such commands, and GMAGIC checks every picture of the file before drawing (a damaged one is refused). The 1982 and 1984 versions fill differently: a file is drawn as **V82** unless it uses commands only the 1984 version has; **D** redraws it in the other version. A file may hold several pictures: **N** (or Space) shows the next one on a cleared page and comes back to the first after the last; **O** draws the next one over the picture shown (the rooms of The Quest and Ring Quest keep an open door or an object as such an overlay). Text is drawn with A2 File Cmd's own font (BOLD.SET), not Penguin's: only a few pictures of the tool's own disks have text, and they differ there. The Quest DR's pictures are 1984 ones that use no 1984-only command: press **D** for them. Recognition reads only the first bytes, so a program that happens to start like a picture is refused by GMAGIC: **H** shows any file in hex. Read only, /RAM kept; the format is described in Graphics Magician pictures.

Malformed pictures are refused. LZ4FH, PRINTSHOP, NEWSROOM, NRCLIP, GMAGIC, FONTVIEW and SHAPES preserve /RAM. Keep both Purplesoft companion files; extended EVE modes require compatible hardware. MACPAINT shows 560 × 192 of its 576 × 720 image; Up/Down pans by 96 lines, **+** / **−** by 48 lines, **T** jumps to the top and **B** to the bottom (lines 528–719). SHAPES shows 24 shapes a page, Space/Down next and B/Up previous.

DGRVIEW also displays complete **80×48 a2dgrx pixmaps**, up to **3,840 bytes**, with one byte per pixel and the original transparency mask. Select the file, use **!** → **Images** → **DGRVIEW**, and enter width **50** (hexadecimal 80). Smaller pixmaps remain centered. **Z** cycles through **×1, ×2, ×4**, using only magnifications that fit within 80×48. The initial view is ×1. **F** chooses the largest fitting zoom (×1, ×2 or ×4) in one step, retaining the chosen checkerboard, mask and orientations. Zoom keeps the original transparency mask and uses the already loaded pixels. **C** toggles a gray/white checkerboard behind the sprite, making transparent pixels distinguishable from opaque black. It also works at ×2/×4; press C again to restore the black background. **M** shows the opacity mask in white on black, including opaque black source pixels. Press M again to restore the colors and chosen background. Zoom works in mask mode; C changes the background restored when leaving that mode. **H / V** mirror the sprite horizontally/vertically, including in zoom and mask modes. Press the same key again to restore that orientation. **O** rotates the displayed pixmap 90° clockwise, including its mirrors, mask and background. Four presses restore its orientation. Zoom decreases when needed to fit; F and Z use the rotated dimensions. A rotation whose height would exceed 48 pixels at ×1 is ignored without changing the view. **R** resets zoom to ×1, rotation, mirrors, colors and the black background. These are display changes; the source pixels are preserved. Saved screen pages and headered DGR pictures keep their original display. Saved screen pages retain their 2,048-byte limit; read/close errors are refused before either screen bank is modified. The viewer preserves AUX `/RAM`; its auxiliary writes are limited to the lo-res screen at `$0400`–`$07FF`.

MacPaint also opens `.PNTG` through Return/I. A renamed ProDOS BIN or typeless MacBinary file is recognized by its `PNTG` type field and actual length, including files whose sizes resemble raw 8K/16K pages. A zero version header alone does not identify a renamed raw MacPaint file: use `.MAC`/`.PNTG` or select MACPAINT explicitly. Full PackBits validation and the confirmation before reusing occupied AUX `/RAM` remain in place.

Return on a Fantavision movie (a BIN at $8400 of 513 to 9,216 bytes, usually named `M.*`) plays it in **FANTA.SYSTEM** (800K and XL), which replaces A2FC while it plays and brings it back after. It plays at the **original speed** (each frame held as long as Fantavision itself takes to draw it on a 1 MHz Apple II). **Tab** switches to the accelerated speed and back, **Space** pauses, **1**-**9** slow the accelerated speed down, **0** removes the delay, **Escape** returns. While paused, **Right / Return** advances one displayed frame and remains paused, in either speed mode. **N** advances through the intermediate images to the next pose stored in the movie, still paused. At the end of a counted movie, or in a one-frame movie, these commands keep the last frame. Space resumes playback; Escape also works while paused. A movie always starts again: a counted one stays two seconds on its last frame, then plays from the start. **S** starts or stops the slideshow: each movie plays once round, stays two seconds, and the next movie of its folder follows (in directory order, the first after the last), until Escape. A movie refused during a slideshow shows its reason three seconds, then the next one plays; a key returns. A damaged tail is cut; a movie with no whole frame is refused with its reason. Ctrl-Reset returns to A2FC. Main memory only: /RAM is untouched. Backdrop: mark (Space) one hi-res picture in the movie's folder before Return; otherwise `NAME` beside `M.NAME`, or else a picture whose name begins with `NAME` (`CHECKERBOARD` for `M.CHECKER`; `NAME` of 4 characters or more), is used. The speed and the slideshow are not kept once back in A2FC.

Return on a Take 1 movie (`MV.*`, Baudville 1985) plays it in **TAKE1.SYSTEM** (800K and XL), which replaces A2FC while it plays and brings it back after. Take 1 movies live on DOS 3.3 disks: open the disk image (`.DSK`, `.DO`, DOS-order `.2MG`) or the real disk from **/**, then Return on the movie; its scenes, actors, backgrounds and fonts are read from the same disk, by name. A movie extracted with **C** into a ProDOS folder plays too, its files beside it under the names extraction gives them -- but ProDOS names keep 15 characters, and on some real disks several files shrink to one name (`BK.UNDERGROUND-1` to `-4`, `SN.HOPPY.WALK.ON1` to `ON3`...): only one can be extracted, and the movie then plays the wrong scene or is refused. Playing from the disk or its image is the faithful way. Ctrl-Reset during a movie returns to A2FC. It plays at the **original speed**, with its fades and speaker sounds; **Tab** switches to the accelerated speed and back, **Space** pauses and resumes. While paused, **Right or Return** advances to the next image page flip and pauses again. Fades run normally before that flip; a scene that waits still needs Return or a paddle button. **Tab** changes speed while paused without resuming, and **Escape** returns. The movie stays two seconds on its last frame and starts again. A missing or damaged file is refused with its name. Read-only, main memory only: /RAM is untouched. The command is limited to 46 characters: an image path of 41 characters at most.

Play music

Music players are included on **800K and XL** and run in the foreground. **P** pauses/resumes, **Escape** stops, and Left/Right selects the previous/next tune of the same type in the directory. A new track starts unpaused. A missing card or invalid data produces an error instead of playback.

MUSIC: MB1 `.MB`, up to 4,096 bytes, six Mockingboard voices. **PT3:** up to 65,535 bytes, three voices or standard two-module `02TS` TurboSound with six. **DUET:** up to 7,168 bytes, type $D5/$D0E7, `.ED`, or compatible BIN names beginning `M.`.

PT3 has two players. A single module of 32 KB or less plays with **GROUiK / French Touch's player** (a 6502 translation of S.V. Bulba's ZX Spectrum player, `A2FILE/PPT3.BIN`), which keeps the whole module in **auxiliary memory**, where ProDOS keeps the /RAM disk. When /RAM holds files, A2FC first asks "ALL /RAM files will be LOST. Continue?" (once while leafing through a folder); an empty /RAM loses nothing and is not asked about. After playback /RAM is rebuilt empty and the message line says "/RAM rebuilt.". With this player the module is read twice, then **the source file is closed before the first note**: the disk may be removed while it plays. Answering **N**, a TurboSound pair, a larger module or a missing `PPT3.BIN` plays with **Vince Weaver's pt3_lib** instead, in main memory only: /RAM is preserved, and the source disk stays open, so leave it mounted. The fourth line of the player screen names the player in use. Both are tuned for the Mockingboard: every period written for the ZX Spectrum's 1.7734 MHz AY is converted to the Apple II's 1.0227 MHz clock, so tunes play at their written pitch (to the rounding of the period). The two players follow their own lineage of Bulba's code and can still differ on some modules: see `src/plugins/ppt3/README.md`.

Mockingboard is detected automatically. A Mockingboard 4c on //c can hide its mouse firmware: A2FC then keeps keyboard navigation and disables mouse calls until that firmware is restored by reset. A plain //c without the card is unaffected. MUSIC and DUET preserve /RAM. Slow reads and large dual modules can delay pt3_lib's playback; 50 Hz is not guaranteed for every tune. Unsupported effects, invalid pointers, runaway command streams and I/O errors stop either player with a message: neither reads outside the module.

DUET uses the speaker when no Mockingboard is present. **1/2** switches speaker/card; **D** cycles speaker pulse width. Keys take effect at the next record on the speaker. Files are validated before playback; FIXTYPES can review old extracted DUET files without changing their content.

Extract archives and run programs

Extract into the other panel

Open the destination first, select the archive and choose its tool in **!**. Return also starts SCIIBIN on `.BSC/.BSQ`, UNSQ on `.QQ/.ACU` and UNWRAP on `.AS` or an AppleSingle file (type $E0, aux $0001); ShrinkIt and Binary II archives and MacBinary files go through **!**. Existing names are not replaced.

UNSHRINK: `.SHK`, stored/LZW1/LZW2 data and disk images. Resource forks, unsupported compressions and members of a type ProDOS cannot hold (a folder type $0F carrying data) are skipped, and the result says so: "N file(s) extracted, M part(s) skipped." **BINARY2:** `.BNY/.BQY`, whose compressed members may need another extraction. **SCIIBIN:** `.BSC/.BSQ` with ordered parts and CRC checks. **UNSQ:** `.QQ/.ACU`, directories skipped and paths flattened. **UNWRAP:** AppleSingle/MacBinary data forks, with omitted resource forks reported.

ShrinkIt uses auxiliary memory, asking first when /RAM holds files, and refuses /RAM as the destination. Escape cancels between blocks. Outputs are read back; errors stop the operation. Cleanup removes only the current new output, where possible; earlier completed files remain. A retained partial file is reported and blocks retry under that name. Preserve it until inspected. Do not assume unsupported or skipped members were recovered; ACU lengths are checked because its stored CRCs are unreliable.

Run a program

Return runs BAS, INT or SYS after confirmation; **X** also runs BIN. The program replaces A2FC and does not automatically return. T lists BASIC without running. 800K/XL include `BASIC.SYSTEM` and `INTBASIC.SYSTEM`; keep the selected program online while its runtime loads. BIN must load from $0800 and end below $BB00. A failed preference save asks whether to run anyway. The confirmation shows A2FC's full return command: from BASIC's] prompt, reinsert its disk and use that command, or `BYE` to the ProDOS selector. Shorten BAS paths over 46 characters if launch fails. Integer BASIC's runtime supports only a subset of DOS commands. Ctrl-Reset in a program A2FC started goes to the monitor's ∗ prompt (memory and /RAM kept), unless the program sets its own reset: BASIC.SYSTEM returns to], FANTA.SYSTEM and TAKE1.SYSTEM return to A2FC.

Work with disks and images

Browse and extract

Return opens supported ProDOS or DOS 3.3 images as folders: PO, DSK/DO, 2MG and supported DiskCopy 4.2 containers. HFS is not supported. Escape at the root leaves the image. **C** extracts the selection or tags to the other panel. An image is read-only: R, K, A, L, D, X, E, W and M are refused. DOS 3.3 panels now support direct T/H/I and Return as described below. ProDOS image files still need extraction with C before viewing. ProDOS extraction supports seedling/sapling files up to 128 KB; enter each subdirectory separately. Existing targets are refused; incomplete outputs are removed where possible and failed cleanup is reported.

A physical DOS 3.3 disk appears in **/** with slot/drive. C extracts to ProDOS on 800K/XL. BIN/BAS/INT headers and sector padding are removed according to the header's exact length; BIN retains its load address, BAS uses $0801. TXT keeps sector data. Leave through **/** before opening a changed disk again.

On real DOS 3.3 disks and DOS-order DSK/DO/2IMG files, T reads text and H reads hex directly; I displays raw BIN pages: $0400 with exactly 1024 bytes in 40-column lo-res, or $2000/$4000 with 8192/8184 bytes in HGR. Return chooses text for TXT, graphics for those picture subsets and hex otherwise. Space pages forward, B back, R restarts and Escape returns. No temporary, writable destination or AUX memory is used. DOSVIEW is included in all ProDOS editions, 800K and XL. Compressed/DHGR pictures and readers without a direct DOS backend still need extraction. Return and I also detect Print Shop 572/576-byte clip art (PRINTSHOP, 800K/XL) and Fontrix fonts (FONTRIX, 800K/XL), plus HRCG BIN fonts named `.SET` or `.FONT` with exactly 768/1024 bytes (800K/XL), alongside the existing direct Print Shop BIN border/font readers (PSBORDER/PSFONT, 800K/XL). They validate and close the source before showing it, without AUX or disk writes. Print Shop fonts no longer need a `FONT.` name: BIN files at `$6000` or `$5FF4`, and ProDOS F5/$1000 fonts, use the same complete glyph-table bounds as PSFONT. Widths, heights, all bitmap pointers and actual EOF must agree. ProDOS F5/$2000 borders with exactly 264 bytes also open through PSBORDER after renaming. BIN borders also open after renaming when their actual length is exactly 144/148 bytes and their load address is `$4800`, `$5800`, `$6800` or `$7800`. The reader shares these checks with identification. Earlier concrete format detections keep priority. For DOS Integer BASIC (`$FA`), use **!** → **Programming** → **DOSINT**; for Applesoft (`$FC`), choose **DOSBAS** (800K/XL): Space/B/R page, go back or restart; Escape exits. It reads directly, without execution, a temporary or AUX. Malformed records, source errors and a line longer than an entire screen page are reported. See direct DOS viewers for bounds and qualification.

For DOS S-C Assembler sources saved as type I (`$FA`), choose **!** → **Programming** → **SCASM**: Space advances and Escape exits. A full validation and successful close precede display; no extraction or AUX.

ProDOS S-C Assembler sources imported intact as BIN also open automatically through Return/I, regardless of their name or load address. Identification checks the first complete tokenized record; SCASM validates every record and closes successfully before displaying the listing. An ordinary BIN is not enough to select this reader. DOS BIN imports keep their existing fallback; direct DOS SCASM still requires type I.

For DOS Merlin/ED-ASM TXT sources named `.S` or `.ASM`, Return and I open **MERLIN** (800K/XL), also available through **!** → **Programming**. The existing reader aligns labels, opcodes, operands and comments and keeps quoted spaces. Space advances; Escape returns. It reads the entire DOS allocation and closes successfully before display, including sectors after the first NUL terminator. It uses neither extraction nor AUX. Sources without these suffixes also open automatically when the first 512 bytes contain at least three complete assembly instructions or numeric equations in Merlin columns. Comments do not count. This is a tentative identification; files with fewer clues keep the generic viewer, and MERLIN remains available manually. Other concrete format detections keep priority. The same rule now applies to ProDOS TXT sources without a suffix, reusing MERLIN's existing reader. The complete source is validated and closed before listing; content identification never executes the source.

Original DOS LISA v2 alternate-B sources with editor header `$1800` also open directly through Return/I in **LISAV2** (800K/XL), available through **!** → **Programming**. They appear typeless in the DOS panel. The reader checks the actual DOS type, line boundaries and mandatory end marker, then reads the remaining allocation and closes before listing labels, opcodes and operands. Space advances; Escape returns. No extraction or AUX is used; later LISA formats use their own readers.

DOS HRCG (`.FONT`/`.SET`), Merlin (`.ASM`/`.S`) and Magic Window (`.MW`) also use the complete 30-character DOS catalog name for detection, even when the panel has shortened it and the suffix is no longer visible. For example, `BLIPPO BLACK.SET` still opens as a font. Trailing spaces and lowercase suffixes are accepted. Existing content-based detections keep priority; the reader revalidates the selected catalog entry and source.

For Music Construction Set exports or editor scores on those DOS disks/images, select the BIN export, main score or `.OBJ` and choose **!** → **Music** → **DOSMCS** (800K/XL). It reads the complete 2304-byte export before playback, with no extraction, temporary or AUX. P/Space pauses, R restarts, +/- adjusts tempo and Escape returns. Paired editor scores are read directly using full DOS names; both files are required. The internal MCSIMPORT/MCSPLAY stages ship beside DOSMCS; see MCS format for bounds.

Add or replace files

IMGPUT: open a ProDOS image opposite a selected source and press C, or choose **!** → **Disks** → **IMGPUT**. It copies only the cursor file, ignoring tags; V refuses. The image needs a free directory slot and enough free blocks. Files above 128 KB and directory growth are unsupported. The source is read once to count it — the length copied is what the file holds, not what the panel showed — then, after the question, the whole image is checked: every block its directories and files name must be marked used in its bitmap, or nothing is written (**Image damaged: nothing written. Run FIXIT.**). That check reads every directory and index block of the image: a second or two on a floppy image, a minute or more on a large one (about 90 seconds at 1 MHz for 16 MB holding 765 files); the activity cell turns and ESC stops it. A `.2MG` whose header says write protected is refused. Data is verified before allocation metadata is installed; interrupted writes can reserve space.

DOSWRITE: with a real DOS disk opposite, C confirms source name, slot and drive. TXT/BIN/BAS/INT up to 65,535 bytes are supported, one cursor file; existing names are refused. The source must be on another device. **DOSREPL** replaces an existing DOS file using free sectors, verifies it, then switches its catalog entry; allow room for both copies. **DOS33W** deletes or renames after confirming the named file and drive. Locked files and inconsistent allocation are refused.

These tools require 800K/XL and standard 35-track, 16-sector DOS disks on a Disk II in slots 1–7. They check physical protection and allocation before writing. A failed write can leave reserved sectors; preserve the disk before further work. Physical VTOC/catalog writes are not atomic.

For a **DOS image**, C supports DSK/DO and DOS-order 2MG. The tool copies the whole image to `A2FC.DOS`, validates and modifies that copy, then installs it with `A2FC.BAK` protecting the old image. Allow space for a second full image. Protected containers and existing recovery names are refused. Keep reported recovery files; the source file is never deleted.

Whole disks, formatting and physical copies

W writes an image to a formatted disk, reads a disk into a new image, or copies disks. PO, DSK/DO and 2MG are supported. Choose the same source/target drive for exchanges. Each prompt identifies SOURCE or TARGET and slot/drive. Writes require **ERASE**, refuse the program disk and verify each block. **F** formats the named drive after ERASE; Escape cancels beforehand. The volume the program runs from is refused, wherever ProDOS lists it, and so is a new name already on line (two volumes of one name cannot be told apart). The disk is identified again just before the first write -- name, size and, without a ProDOS volume, the DOS 3.3 catalog track and the CP/M directory -- and a changed disk is left alone. Disk II formatting clears / RAM, asking first when it holds files, and refuses to run from /RAM; a write-protected floppy fails before AUX is used, /RAM intact. A Disk II format then reads all 280 blocks back; other devices read back their boot block and volume header only, and the message says which.

NIBCOPY copies standard Disk II tracks with one or two drives at normal 1 MHz speed. Write-protect the source first. Accept AUX use only after saving /RAM, then confirm destruction of **all target files, including locked files**: the confirmation names the target's ProDOS volume, and the program's own volume is refused. Single-drive mode requires target confirmation after every exchange. Each source track is read twice and the written track verified; before each later write the target must still hold the previous track, or the copy stops (see NIBCOPY.md for what this does not catch). Escape stops between tracks. An incomplete target is not a valid backup. Nonstandard/protected, 13-sector, half-track and weak-bit formats are unsupported; this is not a flux copier.

PASCAL and **CPM** extract supported images into the other panel without modifying the source. Names are adapted to ProDOS; existing targets are skipped. **SMALLDOS** (Disks menu) extracts Wilderness catalogues from raw 35-track `.DSK`/`.DO` images in DOS order or `.PO` images in ProDOS order. Select the image with a ProDOS destination folder opposite. Both reversed game tracks and normal demo tracks are supported. All allocated sectors, including padding, are kept because the catalogue records no exact byte length. Type A becomes Applesoft `$FC/$0801`; type B becomes binary `$06/$0000` because its load address is unknown. Names are adapted to ProDOS, collisions are skipped, and every output is closed and compared with the source before being counted. Escape stops extraction; files already verified are retained. An incomplete current output is removed, or its failed cleanup is reported. JOURNEY has no catalogue and is refused. WOZ/flux images and SMALLDOS writing are unsupported; the source image and auxiliary memory are never written. Pascal text retains its header/padding. CP/M data is rounded to 128-byte records; text ends at $1A. Unsupported directory layouts are refused. **PASCALW** and **CPMW** do the reverse: every file of the other panel's ProDOS directory goes into the selected Pascal or CP/M image, a name already there being skipped. PASCALW writes only after the last block in use (Krunch a volume whose free space is all between files); CPMW takes files up to 16 KB and refuses an empty volume, whose sector order it cannot establish. **DISKCMP** compares two volumes or images block by block, read only. Use **IMGCONV** for container conversion, not extension changes. No physical-write operation promises rollback through a power failure.

Find and maintain files

Press **!**, choose a category, then a tool. Return opens; Escape goes back. Up/Down moves one entry, Left/Right six, and an initial jumps by name. Categories describe tasks; no category disk swap is needed. Except COMPARE, VERIFY and MOVE, the tools below require **800K or XL**. Read each result, especially partial scans, skipped files and failed cleanup.

Files and navigation

- **COMPARE / CRC / VERIFY:** compare opposite namesakes byte for byte; calculate CRC-32; read files or volume blocks and report errors.
- **TAGPAT / SEARCH / FIND:** tag by pattern; search current-directory text; search a volume by name or contents (" prefix). FIND: TAB filters, N pages, Return jumps and V shows text matches; a directory it could not read ends the results with "some paths skipped" instead of "complete".
- **GOTO:** P opens a path; A adds a favourite, D + digit removes, M + two digits reorders, 1–9 jumps.
- **FAVRUN** (Programming menu): select a SYS, BIN, BAS or INT file and use A to save its absolute path. 1–9 launches a favourite through RUN, with the usual confirmation; D + digit removes it, M + two digits reorders. Up to nine paths (63 characters each) live in `A2FILE/FAVRUN.CFG` next to the settings file. Missing or unreadable programs are refused. Saving uses exclusive `FAVRUN.TMP` and recoverable `FAVRUN.BAK`; a leftover must be reviewed before saving again. Power loss during replacement is not atomic.
- **RENAME / DATE / TXTCONV:** batch names; modification dates; CR/LF/CRLF, high-bit, tabs or UTF-8 accent conversion. Conflicting names are preserved.
- **IDENT / FIXTYPES:** identify formats; confirm attribute changes individually. Hints do not prove a damaged file is valid.
- **SYNC / TREE:** copy missing/newer files recursively; show the directory tree, read only, in disk order, with file types and directory sizes.

TAGPAT uses = for any string and ? for one character; comma filters include `T04`, `>2000`, `<2000` and `D` (today). T adds tags, U removes, X replaces. GOTO paths are at most 63 characters. DATE accepts `DDMMYYYYHHMM`, 1940–2039; creation dates remain unchanged. SYNC preserves destination-only and newer files, refuses overlapping trees and stops if replacement/recovery fails. An unknown source date does not authorize replacement. Keep `A2FC.SYNC/A2FC.BAK`. TREE counts a forked (GS/OS extended) file as a file, with the size its directory entry gives, as the panels show it; its forks are not opened. At the top of a volume, TREE marks `<- boots` on the first SYS file whose name ends in `.SYSTEM`: the one ProDOS 8 starts. An unknown or forked entry before it leaves the volume unmarked.

Volumes and recovery

VOLINFO audits space, allocation and fragmentation: M bitmap, F file blocks, E export, N/P pages. Partial counts are unconfirmed; only a complete export ends with `END REPORT`. F and E leave the audit's figures as they are: a read error or a full destination while listing a file spoils that listing or report, not the audit. **VOLNAME** renames a volume and updates affected paths; a name already online is refused. **MKIMAGE** creates an empty PO/2MG data volume, without a boot program, up to 32,767 blocks. It closes and reads back the whole image before reporting success; ESC also cancels during verification. Failed outputs are removed when possible; a cleanup failure identifies the incomplete file retained on disk.

FIXIT only reads. It reports ProDOS directory/allocation faults and refuses incomplete scans rather than declaring lost blocks. For volumes above 4,096 blocks, Q checks directories only; F performs a full scan, which uses AUX: if /RAM holds files you are asked first (an empty /RAM is not asked about), and /RAM is rebuilt empty afterwards, or after a Ctrl-Reset during the scan. A quick clean verdict is not a full allocation check. It does not check DOS disks. When two drives carry the same volume name, open it from the volume list (`Two volumes named /X: pick it in the volume list.`).

REPAIR shows a plan and requires **F**, then **FIX**, before writing. It refuses the program volume, cross-links and uncertain scans. Unclaimed (lost) blocks are given back only when nothing else is wrong with the directory tree: beside any other fault they may belong to a file a damaged pointer no longer reaches, and REPAIR then writes nothing at all (`Lost blocks may hold a damaged file: nothing written. See FIXIT.`). If the tree claims blocks the bitmap calls free, REPAIR says so: copy the files to another volume and write nothing to this one, because the next allocation can land on a file. Even alone, lost blocks are listed by number on the plan screen and need a second word, **FREE**, before **FIX**: a file pointer moved onto a block that was already lost looks exactly like this. If any file is missing or short, copy what still reads before freeing anything. A directory REPAIR cannot trust (a key block that is no directory header, a volume directory out of the place its header gives, a wrong back link in a subdirectory) stops it with `Directory not trusted`. Writes are read back, restoration is attempted on error, and the volume is rescanned. Backups are only in RAM: interruption can leave a partially applied plan. Large scans can take minutes. REPAIR cannot see a volume header whose block count was lowered: it then clears the bitmap bits past the new end.

DIRSORT changes the order of the entries of the directory the active panel shows, on the disk itself: the order ProDOS searches, and the one it boots by (ProDOS 8 starts the first SYS file whose name ends in `.SYSTEM` in the volume directory, marked `<– boots`). Space carries the entry under the cursor, Up/Down move it, Space drops it; N sorts by name, T by type then name, D by modification date and time (oldest first; undated entries first) then name; R returns to the disk order; W writes after a confirmation that names the directory; Escape writes nothing. Only the order changes: deleted entries keep their slots, the header and file count are untouched, and a subdirectory that moves has its header's parent link rewritten. DIRSORT refuses an image, the volume list, a volume whose name is on two drives, a directory of more than 4 blocks (51 entries), a write-protected directory or volume, an order that would move a locked subdirectory (its header would be written), and anything FIXIT would question (file count, block links, storage types, a subdirectory that does not point back at its entry). Just before writing it reads everything again and writes nothing if the disk changed. Each block written is read back; on a failure what was written is put back and the message says so, or `RESTORE FAILED (block N)` if that fails too. A subdirectory header is checked by its link and a checksum, its original not being kept in memory. A power cut during the writes can leave an entry twice or missing: back the disk up first.

UNDELETE recovers validated deleted files to another volume without changing the source. Reused/ambiguous allocation, deleted directories and resource forks are refused. **RESCUE** retries reads and zero-fills unreadable chunks, recording losses in `BASE.LOG`. File output is `BASE.REC`; volume parts `BASE.P01`, `.P02`, etc. must be concatenated in order. Partial output remains. Inspect the log and recovered contents: neither tool guarantees recovery of missing data.

WIPE F checks allocation before zeroing free blocks; **W** destroys an entire volume after ERASE, excluding the running program's volume. Both read the volume header before the question, which names the volume found there, and compare it again after the answer (F once more after its check): a disk changed meanwhile, or one that no longer reads, gets "Disk changed or unreadable: /NAME. Nothing written." A disk swapped once the writes have begun is not detected. **BOOTBLK** checks the volume names on the disks before its question and compares block 2 of target and source again after it (same message), then copies boot blocks, with readback and RAM-only originals for attempted restoration. **BLKVIEW** reads blocks and can extract a range; **BLKEDIT** changes bytes of one block and writes it back after ERASE, then reads it back (never on the running program's volume, and not on a disk whose block 2 no longer has the CRC-32 it had when opened; an image of one or two blocks is checked on its block 0); **DISASM** reads 6502/65C02 instructions and exports text. Their exports can remain partial after an error. None of these tools should be tried first on an irreplaceable disk.

Two volumes with one name. A path names a volume, not a drive: when two drives on line carry the same volume name, ProDOS opens a path on one of them, which may not be the one a tool would write by its drive. WIPE, BOOTBLK, BLKEDIT, RESCUE, VOLINFO, FIXIT and REPAIR then refuse a volume given by the panel's path: "Two volumes named /X: pick it in the volume list." A row of the volume list is opened by its drive and still works. BLKVIEW and UNDELETE refuse with their usual message (UNDELETE from the volume list too, since it reads the directory through its path); BOOTBLK refuses when the volume it was started from has a twin; MOVE copies and verifies through ProDOS instead of moving entries in place. Rename one of the two (VOLNAME) to work by path again. DIRSORT works by path only and refuses such a volume.

Limits, troubleshooting and VDrive

Read the status before acting

Long copies distinguish **Copying** from **Verifying**; the verification counter starts again at zero. Every long tool shows a bar or, where none fits, turns a / \ mark at the end of the line above the messages (DOS3.3: the last cell of the key bar, which also turns inside DOS's own long calls: the format, and the retries on a diskette that was never formatted). A drive change or a head recalibration can still hold it for a second or so. Do not remove media just because a counter has stopped; record the screen and wait for a result or a safe cancellation point.

Escape cancels where the active tool permits it. Completed work remains and partial output may require recovery. A result that reports skipped, unsupported, unreadable or retained files is not a fully successful batch. Check the actual message before retrying or cleaning up.

Symptom	Action
Startup refusal	Choose the matching CPU/ROM edition; ProDOS needs 128 KB and 80 columns.
Missing/stale tool	Use XL for tools omitted from 800K; keep matching program/plugins.
Old catalog or a large directory	Ctrl-R rereads; cross window edges beyond 139 entries. Large windows use disk order; inside an image only the first entries show.
Deep traversal refused	Work on smaller subdirectories; an incomplete tree must not be deleted.
Temporary/backup blocks retry	Preserve candidates and follow incident recovery.
Run or image opening fails	Check paths, runtime, type and actual container; renaming extensions does not convert.

Boundaries to keep in mind

A matching size/date does not prove a recovered file complete: COMPARE checks bytes, M only compares size/date. Unsupported forks or storage may be refused or skipped with a report. AUX consent clears /RAM without backing it up. Verification cannot make physical metadata writes atomic across a power cut.

VDrive: remote volumes

A detected Super Serial Card or //c serial port can expose two remote ProDOS volumes at **115,200 bps**. VDrive tries slot 2 (the //c's modem port) first, then slots 3 to 7, and **never slot 1**: that is the printer's (a //e's printer SSC, the //c's printer port), and VDrive would reprogram it and send it its packets. It does not even read that card. A VDrive cable on a slot-1 card is therefore not served: put the card in slot 2. In slots 2 to 7, a Super Serial Card whose mode switches (SW1-5 and SW1-6) are not set to communications mode -- printer mode, or a SIC emulation -- is a printer's: VDrive reads those switches first and skips the card without writing to it. Set the card that talks to the VDrive host to communications mode. A //c has no switches, so nothing tells a printer on its modem port from a modem: port 2 is always taken when present, and a printer plugged there receives VDrive's packets. Keep printers off the //c's modem port while A2FC runs. Use ADTPro's virtual-drive server, `veserver.py` or `surl-server`. The two volumes take the first slot with no unit of its own in the ProDOS device list; the status line identifies the serial interface and that slot. When every slot already has a unit (a //c or a IIe whose SmartPort fills them), nothing is installed and the status line says `VDrive: no free slot.`; nothing is installed either when ProDOS has no room left for an interrupt handler. Browse and copy as with local volumes. The host supplies date/time during reads. A disconnected server produces an I/O error; it is not an empty or absent file. A reply that comes too late is drained before the next block, so one slow answer costs one I/O error, not every call after it. After a failed read the buffer holds whatever had arrived, as with a disk drive; ProDOS treats it as undefined. The driver is removed on quit or program launch.

The 6551 of a Super Serial Card or a //c holds one received byte: at 115,200 bps a byte arrives every 88 cycles, and VDrive's receive loop takes 33 cycles a byte plus 19 to see the next (measured in the simulator, both editions). VDrive has run in the POM2 emulator only, whose serial card buffers 4 KB: it has **not** yet been tried on a real Super Serial Card or //c port.

For reproducible failures, note the A2FC version/edition, machine, disk type, full paths, exact message and actions. Keep a disk image before any repair. The project's issue tracker and detailed data-safety notes are linked in the last chapter. Automated emulator tests do not replace trials on physical drives.

Credits and further reading

Additional readers launched from ! are described in Legacy readers. This includes Music Construction Set Mockingboard exports and paired editor scores: select the export or either file of a NAME + NAME.OBJ pair, choose Music then MCS. P/Space pauses, R restarts, +/- adjusts tempo, Escape returns. Both score files must be in the same ProDOS directory; see MCS format for naming and compatibility limits.

A2 File Cmd is **GNU GPL v3** software by **Arnaud Verhille**. Its two-panel interface draws inspiration from A2Command and Norton Commander. Code provenance, source notices and redistribution terms remain in the repository and LICENSE.

Acknowledgements

- **cc65 / Oliver Schmidt:** toolchain and Apple II startup sources.
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- **Jerry Hewett and Gary Desrochers:** public-domain Hyper-FORMAT routines, carried by ADTPro and adapted for Disk II formatting.
- **Apple ProDOS references:** filesystem, MLI, paths and allocation rules.
- **Paul Lutus:** Electric Duet format and original player; **Alex Patalenski** and **Emil Dotchevski:** improved speaker player and its published listing.
- **Cybernesto, electric-mock:** DUET's Mockingboard rendition; **Vince Weaver, pt3_lib:** PT3 player work adapted by A2FC (fallback).
- **GROUiK / French Touch (2019):** the 6502 PT3 player A2FC plays first (GPLv3, from the *One More Thing* demo sources); **S.V. Bulba:** the ZX Spectrum Vortex Tracker II player it translates; **Ivan Roshin:** its note and volume table generators.
- **Andy McFadden, CiderPress II:** format notes and real reader test samples.
- **NuLib / ShrinkIt references:** NuFX and Binary II specifications.
- **David Schmidt, ADTPro:** disk-transfer and virtual-drive workflows.
- **POM2 / Arnaud Verhille:** separate emulator for repeatable tests.

The complete attributions, reference URLs and source-file notices are preserved in Credits and provenance. Keep that document, the affected source notices and LICENSE when redistributing a modified build. DEMO/README on the XL disk tells generated demonstrations from the CiderPress II samples it reuses; data/CP2/README.TXT describes that corpus.

Where to go next

- **After an incident:** open RECOVER on the program disk and follow this guide's recovery chapter before retrying or deleting anything.
- **Failure coverage:** Data safety describes temporary names, retained originals, error handling and physical-write limits.
- **Real machines:** Hardware checklist covers disposable test media and records the actual machine/drive results.
- **DOS3.3 internals:** Developer guide holds implementation, build and timing details removed from this concise user guide.
- **Media:** Sample media documents test material and sources.

- **Builds and extensions:** the repository README, `sdk/README.md` and `bench/README.md` describe development and test workflows.
- **Bugs and source:** A2 File Cmd on GitHub.

This is the guide for **0.9.9**. It describes supported behavior and limits; it is not a claim that physical hardware qualification has finished. Keep original disks safe, use matching program/tools, and read each operation's result before starting the next one.

Direct Newsroom panels and layouts

Return/I use IDENT on both ProDOS and DOS 3.3. PN. files (BIN $4000) open NEWSPAN: recovered text and embedded photo references. PG. files (BIN $98A5) open NEWSPAGE: letter/legal format, banner choice, page name and component slots. Original fonts, placed photos and complete newspaper-page assembly remain unsupported. PH./BN. retain their bitmap viewer; DOS originals use DOSNEWS. The source and AUX remain untouched. Space advances, Escape exits. Epistole/DOCVIEW and Extasie remain ProDOS readers, never DOS 3.3 readers.

MultiScribe and Apple Writer

MULTISCR reads the recognized MultiScribe TXT/WPF variants as paged plain text. The same documents imported intact as ProDOS BIN also open directly through Return/I: identification uses the existing format headers regardless of filename, then the reader validates the complete document. Fonts, styles and ruler records are validated; original proportional typography is not reproduced. **APPLEWR** reads Apple Writer text in ProDOS and directly on DOS 3.3 disks/images. It supports `.LM`, `.RM`, `.PM`, `.CJ`, `.LJ`, `.RJ`, and approximates `.FJ` as left alignment and `.FF` as a blank separator. Width is limited to 79 screen columns; unsupported or out-of-range commands stay visible. It never executes WPL, includes or mail merge. Both use MAIN only and preserve `/RAM`. Space/Down advances a page, Escape returns. Return automatically chooses these readers for their recognized header/initial layout-command candidates; `!` allows explicit selection. See format limits.

Print Shop borders and fonts

PSBORDER previews the three stored 24×14 border patterns. It does not assemble a frame or apply the four-byte option prefix. **PSFONT** previews individual glyphs: Left/Right or Space changes the glyph, Up/Down scrolls its rows, Escape returns. Space and the external `@` placeholder have no bitmap preview. Both accept original BIN files directly in ProDOS or DOS 3.3, validate complete input and use MAIN only. Return/I recognizes borders with matching exact length/load address and fonts with matching glyph-table bounds, including renamed files; use `!` for manual selection. ProDOS `$F5` aux $2000 borders (264 bytes) and aux $1000 fonts (95 entries) are also supported; frame assembly and masks remain uninterpreted. See format and qualification.

Magic Window and LISA v3

MAGWIN reads original Magic Window BIN documents after their 256-byte header, in DOS 3.3 or ProDOS. High ASCII becomes readable text; printer controls stay visible as `^A`, `^[`, etc. The header/footer and printed styles are not rendered. Return recognizes `.MW` candidates with a matching header; `!` also permits other names. Limits: 48 KiB, validated before display.

LISAV3 lists standard ProDOS LISA v3 sources (INT or imported BIN, aux $1000–$3FFF), recognized automatically through Return/I, with up to 512 packed labels, instructions, operands, macros and comments. Lengths, line boundaries, strings and symbol references are checked before display. The known incompatible `ANIX.EQUATES` dialect and LISA v4/v5 are refused. Space/Down advances a page; Escape returns. Both use MAIN only and preserve `/RAM`. Qualification and remaining limits.

DOS fixed-record text (DOSREC)

Select a DOS 3.3 text file and launch **DOSREC** through ! → Files. Enter the record length in decimal (1–4096); an empty Return selects 256. DOS does not store this length or an exact text EOF. The displayed extent ends at the highest allocated sector and includes sector padding.

N/P select the next/previous record; Space/B page within a record; R returns to record zero; L changes the length; ESC returns. Offsets are hexadecimal. −− marks an unallocated sector (~ in ASCII), while `00` is an allocated NUL byte. Control bytes appear as dots, high-bit text is shown as ASCII. See DOS-RECORDS.md for validation and format limits. Skipped T/S ranges are accepted only by this inspector; logical positions are bounded to 560 sectors. Sources and `/RAM` are preserved.

Print Shop ProDOS $F5 aux $2000 borders (264 bytes) and aux $1000 fonts (95 entries) use PSBORDER/PSFONT automatically, including renamed files. The border preview shows three motifs; frame assembly and masks remain uninterpreted. Space and the external `@` graphic have no displayed bitmap.

LISA 8/16 v4/v5 (LISAV4)

Return recognizes ProDOS INT sources with aux $4000–$5FFF and a plausible LISA 8/16 header. Use ! → Development → LISAV4 to choose the reader manually. It lists variable-length symbols, macros, 24/32-bit numbers, address modes and comments, using the tab stops saved in the file. Space/Down advances a page; ESC returns. Numeric 32-bit decimals are unsigned.

The complete source is validated and closed before display. Sources and `/RAM` are preserved. The symbol table is limited to 2816 bytes and the file to 65535 bytes; malformed or unsupported tokens are refused. See LISA v4/v5 qualification and limits.

PFS:Write documents (PFSWRITE)

Return opens qualified ProDOS PFS:Write documents (type $16, aux $0002). Use ! → Files → PFSWRITE for manual selection. Space/Down/Return advance, ESC returns. The reader skips page/printer settings, shows plain text and exposes embedded controls as `^A`, `^I`, `^L`… Formatting and database merges are not interpreted. File/Plan data and PFS program modules are refused. Complete body/EOF checks and a successful close precede display; source files and AUX /RAM remain intact. Maximum body: 64,511 bytes. See format, corpus and limitations.

PFS:File databases (PFSFILE)

Return opens ProDOS $16/$0001 A2CD00 databases, or use ! → Files → PFSFILE. The viewer lists active records with form labels and multiline values. Space/Down/Return advance; ESC returns. It validates the complete file I/O and all reachable form/record chains before displaying content. Sources and AUX /RAM are preserved. Limits: 256 KiB, eight cells per form/record, 32 form labels. Reports and older custom disk formats remain unsupported. See structure and qualification.

PFS:Plan spreadsheets (PFSPLAN)

Return opens ProDOS $16/$0004 B00 sheets, or use ! → Files → PFSPLAN. The viewer lists row and primary column labels with stored decimal values; blank cells remain visible and formula results are marked. Row/column formulas are listed separately without evaluation or recalculation. Space/Down/Return advance; ESC returns. Full validation and successful close precede display; sources and AUX /RAM are preserved. Limits: 32 rows, 16 columns and 512-byte label/formula blocks. Group headings, print formats, per-cell formulas and other profiles remain unsupported. See structure and qualification.

Formats selected for 1.0

WORDPERF, MOUSEWR, BSFILER and MULTPLAN display text, database records and stored worksheet values. MVMOVIE gives a 40×48 colour frame preview. ProDOS BIN/$6000 or imported BIN/$0000 films can be opened with Return or I even after renaming: bounded shape/bitmap headers and the actual file length identify a candidate, then the reader validates the entire film before display. This preview uses MAIN only and preserves `/RAM`. Space/Return/Down/Right advance. **S** starts automatic frame advance; S again pauses on the current frame. **1 / 2 / 3** select fast/normal/slow advance, during playback or while paused, without advancing a frame. A speed change keeps auto active and restarts the current wait. Other keys stop auto and act as usual. The delays are about 0.7/2.9/5.8 seconds at 1 MHz, plus rendering time (faster on an accelerated machine). Normal is the default; navigation retains the selected speed and reopening restores normal. **Q** toggles automatic looping without advancing or stopping playback. With looping enabled, auto rereads the checked animation stream after the last frame and displays the first again, retaining speed, background and mirrors. S still pauses/resumes; Q again restores return to the panels at the end. Manual next at the last frame returns normally. Reopening starts in manual mode with looping disabled. B/Up/Left return to the previous frame, R restarts, **L** jumps to the last frame and Escape exits. L uses the frame count from complete initial validation and rebuilds the last frame without displaying intermediate ones. **+ / −** advance/back up ten frames, stopping at the last/first frame. These jumps also skip intermediate rendering and retain B/R navigation. On a long film, this can require rereading many commands. **O** toggles the background alone or the preview with actors, on the same frame. The choice persists during navigation and resets when reopening a film. Redrawing uses checked bitmap reads and preserves the source file. **H / V** mirror the complete preview horizontally/vertically, including the background-only view. Toggle again to restore either orientation; the choice persists during navigation and resets when reopening. Previous at the first frame keeps that frame. The preview remains opaque and reduced to 40×48; auto does not reproduce the original HGR animation timing. DGMAGI draws DHGR .DPC pictures with a substitute font. Renamed ProDOS BIN/$4000 pictures also open through Return/I when valid sampled commands include drawing and coordinates beyond column 279. Narrow pictures remain ambiguous and retain their existing routing; select DGMAGI manually when appropriate. PCSVW gives a static .PB preview with library outlines. **O** toggles between that preview and the original bitmap background, without rereading the file. ProDOS BIN/$4000 PCS tables also open through Return/I after renaming when the saved object table, sampled polygon bounds and actual file length agree. The reader validates the complete file before requesting AUX consent. Graphics may erase all files on /RAM only after confirmation, and reject source files located there. See V1-FORMATS for supported versions, extraction requirements and outstanding 800K packaging. These readers do not execute document code. MOUSEWR also accepts intact MouseWrite imports as ProDOS BIN/$0000, through Return/I or manual selection. The supported MW versions 1/4/5, header flags, width 1–120, complete 512-byte header and complete rows remain mandatory. The entire document is read, validated and closed before display; MAIN-only text preserves `/RAM`. Other BIN auxiliary types retain their fallback. MULTPLAN also accepts normal 1.06 worksheets imported intact as ProDOS BIN, with automatic Return/I based on the existing eight-byte signature. The reader still checks the complete workspace, cell pool, record pointers and decimal values before display. Cached results and formula tokens are shown without evaluation; Symbolic saves and direct DOS R remain unsupported.